

Gestures for Blind People to Specify Spatial Focus of Vision Language Models on Head-Worn Devices

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Abstract

Head-worn visual assistance can make it difficult for blind users to obtain descriptions focused on the part of a scene they want to examine. Thus, we propose two interactions for head-worn devices that enable blind users to direct vision-language models (VLMs) towards a desired area of the scene using hand gestures. In a study with five blind participants, we compared two focusing methods for head-worn devices, a diagonal-framing method and a fingertip-cursor method, with smartphone-based image capture. The diagonal-framing method tended to achieve higher precision across targets, while also imposing higher interaction workload. Participants mentioned that they were able to obtain more relevant information than with the smartphone-based method.

CCS Concepts

• Human-centered computing → Accessibility design and evaluation methods.

Keywords

vision-language models, head-worn devices, blind people

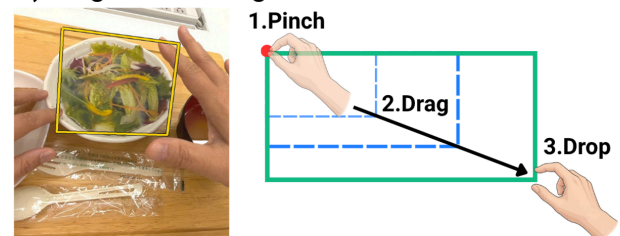
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1 Introduction

VLM-based visual assistance using wearable devices, especially head-worn devices (HWDs), helps blind people obtain information through an egocentric camera with hands-free interaction [2]. However, prior work has shown that blind users cannot always properly frame an intended object with an egocentric camera [3], resulting in frames that include many irrelevant objects. Although users can

A) Diagonal-framing method



B) Fingertip-cursor method

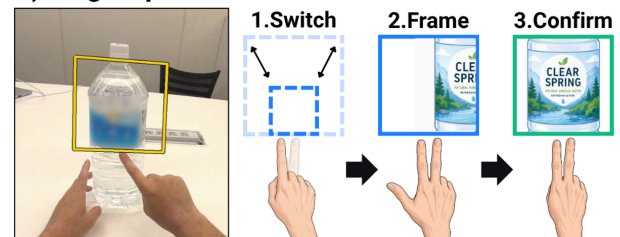


Figure 1: A) the diagonal-framing method, where users pinch, drag, and drop to specify a rectangular region. B) the fingertip-cursor method, where users adjust the region size with the index and middle fingers, fix it as a frame by opening their thumb, and confirm by closing it.

repeatedly verbally query VLM-based visual assistance to obtain the desired information [5], doing so can be burdensome.

Researchers have explored hand-based interactions on wearable systems that let blind users direct visual assistance toward specific objects. For example, Lee *et al.* used the position of the user's hand in the egocentric view to localize an object of interest [3], while TouchScribe uses interactions such as holding, touching, and pointing to describe the object being interacted with [1]. However, these approaches may not sufficiently support users when the information they want to know concerns objects they cannot touch (*e.g.*, food on a plate in a restaurant) or hold (*e.g.*, a poster), or spans a larger region (*e.g.*, an entire row on a shopping shelf).

Thus, we explore an interaction for VLM-based visual assistance on HWDs that lets blind users adjust both the location and size of their region of interest using hand gestures. We compare two



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Figure 2: Targets in the user study.

gesture-based methods for HWDs (Fig. 1) to an existing smartphone-based method. We evaluate the accuracy of capturing an intended part of the scene and the workload associated with each method.

2 Implementation

We implemented our system on Apple Vision Pro, using ARKit for main-camera and hand tracking. The images from the main camera were fed to GPT-5.6 Terra for visual description, and responses were read aloud via speech synthesis. We designed two gesture-based methods that allow users to focus VLM-based visual assistance on a specific spatial region, as described below (Fig. 1).

Diagonal-Framing Method. Users define a target region by performing a pinch gesture with their thumb and index finger. The starting and ending positions of the pinch interaction define the rectangular region fed to the VLM. For confirmation, a short sound is played when the interaction begins and ends, and a continuous sound is played during the interaction.

Fingertip-Cursor Method. Users control the size and position of a square region placed on the tip of their index finger using a pointing gesture. Pointing with the index finger will define a 15 cm × 15 cm region, and also extending the middle finger will make it a 30 cm × 30 cm region. Opening their thumb will confirm the region size, and closing it will capture the selected region to initiate inference.

3 User Study

To investigate whether our proposed interaction methods enable users to capture their intended area of focus, we conducted a user study with five blind participants (3 male, 2 female; aged 22–65). We compared our two HWD interactions and a conventional smartphone application called Seeing AI [4]. After a training session, participants were instructed to capture specific objects while excluding the irrelevant surrounding area as much as possible. The target objects were a label on a plastic bottle, a salad placed on a tray, and an A3 poster attached to a wall (Fig. 2). Each participant completed three trials for each combination of condition and target, resulting in 27 photos per participant (3 conditions × 3 targets × 3 trials), with the order of conditions counterbalanced.

We measured *Precision* as the proportion of the focused area occupied by the target. For the two HWD interactions, we also measured *Recall* as the proportion of the target contained within

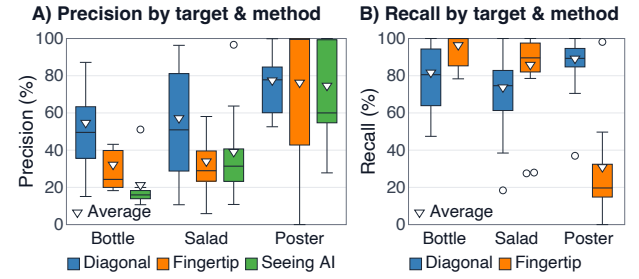


Figure 3: Precision and recall by method and target.

the focused area, only when its main camera captured each target's full extent. Finally, we conducted a post-interview and obtained the workload measures (Raw NASA-TLX), where lower scores indicate low workload, for each condition.

4 Results

The diagonal-framing method achieved consistently higher precision across the three targets than smartphone-based capture, while also maintaining relatively high recall across target sizes (Fig. 3). In contrast, the fingertip-cursor method showed target-dependent behavior. For the bottle and the salad, it captured most of the target, as reflected in its high recall, but also included more surrounding content, which lowered its precision. For the poster, the fingertip-cursor method often failed to cover much of the target even when enlarged to 30 cm × 30 cm, which reduced recall.

The mean Raw NASA-TLX score was 5.80 (SD=1.66) for the diagonal-framing method, 4.83 (SD=1.53) for the fingertip-cursor method, and 2.83 (SD=1.80) for Seeing AI. Both proposed methods imposed a higher workload than smartphone capture, with the diagonal-framing method being the most demanding. Participants attributed the low workload of Seeing AI to their familiarity with smartphone interactions compared to HWD interactions.

All participants valued the region specification approach over capturing a full image. For example, P5 mentioned that our approach allows them to focus on the object of interest while saving time, whereas Seeing AI's descriptions were broad and expressed uncertainty about whether it was about the target object. P3 noted that pinpoint control would be useful when locating a specific product while shopping, and P2 mentioned that gestures let them specify an area of interest more easily than they could through verbal instructions. P3 and P4 also suggested usage in identifying specific parts of vending machines or control panels.

5 Conclusion and Future Work

Our study showed a trade-off between performance and workload. The diagonal-framing method provided more precision and flexibility, but at the cost of a greater workload, whereas the fingertip-cursor method had lower interaction demands while offering less consistent performance. Importantly, all participants recognized the value of our approach. Future work will focus on reducing this workload by incorporating the user feedback from this study.

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